

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2458159.77502 +/- 0.00047 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.086 +/- 0.026
Transit depth (Rp/Rs)^2	0.74 +/- 0.45 [%]
Orbital Inclination (inc)	81.57 +/- 2.79
Airmass coefficient 1 (a1)	1.312 +/- 0.024
Airmass coefficient 2 (a2)	-0.1951 +/- 0.0047
Scatter in the residuals of the lightcurve fit is	1.75 %
Best Comparison Star	None
Optimal Aperture	5.27
Optimal Annulus	8.11
Transit Duration (day)	0.1149 +/- 0.0088

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.086 $\pm$ 0.026 vs 0.1178 (deviation 0.74 σ)
Duration fit vs published	0.1149 d vs 0.125 d (deviation 0.7 σ)
Mid-time O-C	1570.3 min
Depth SNR	2.0
Residual scatter	1.75 %

Light curve

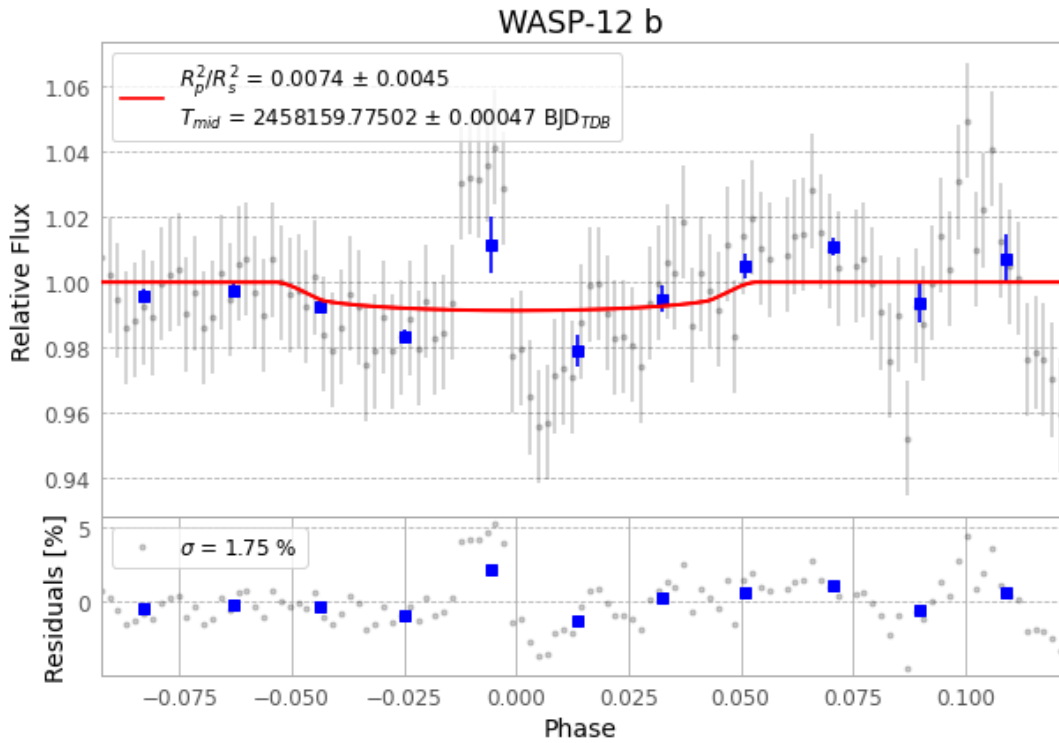


Figure 1 — FinalLightCurve_WASP-12 b_2018-02-09.png (SHA-256 39b29c239aab5918...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [347, 270], 1 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 dd01a78f13566549...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit b365494

Data provenance

114 FITS frames (75 MB), WASP-12180210040314.FITS → WASP-12180210094212.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-12-180209.log (37c7b2e58a3d...), FinalLightCurve_WASP-12 b_2018-02-09.png (39b29c239aab...), FinalLightCurve_WASP-12 b_2018-02-09.csv (6840820dccf9...)

Compute resources

Wall time 95.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-22 09:35Z.